

WINNER ENGINEERING

LEARNING HUB SERIES
MECHANICAL VENTILATION
DUCT SYSTEM

Topic

1. Duct Insulation
2. Duct Accessories
3. Duct System Connection



Duct Insulation

Reason for Duct Insulation

1. Thermal
2. Acoustic
3. Weatherproof
4. Fire Rate Requirement



THERMAL EXTERNAL INSULATION

INSULATION THICKNESS FOR DUCT IN DIFFERENT CONDITIONS

TYPE	CONDITION	INSULATION THICKNESS (MM)
SUPPLY	NO AC SPACE	50
SUPPLY, FRESH AIR	AC SPACE	25
RETURN, EXHAUST	NO AC SPACE	25
SUPPLY, RETURN, EXHAUST	EXPOSED TO WEATHER	50 / HYBRID PLASTER
SUPPLY, RETURN, EXHAUST	BELOW THE ROOF	50

ACOUSTIC INTERNAL INSULATION

- Perforated duct sheet at internal lining
- Dampens sound reflection & turbulence within
- Modelling thickness 50mm
- Unless otherwise stated, length to be 5m from AHU & 3m FCU



ACOUSTIC INTERNAL INSULATION

- Is pre-insulated duct considered internal insulated?
- What is the thickness for modelling consideration? 20mm



WEATHERPROOF INSULATION

- GI /Aluminium Jacketing
- Hybrid Plaster

20mm



FIRE REQUIREMENT

- Fire Rated Board
- Fire Wrap



FIRE RATED DUCT

- Fire rated board
 - 2-Hour - 75mm thickness
 - 4-Hour - 150mm thickness
- Fire Wrap
 - 2-Hour - 28mm thickness
 - 4-Hour - 28mm thickness



FIRE DAMPER OR FIRE RATED DUCT

Common function but different context

ITEM	FIRE DAMPER	FIRE RATED DUCT
Construction	Follow structural thickness	Thickness based on rating, fire rated board, or fire wrap
Usage	Floor / floor	Duct length
Function	To block fire from spreading between separate compartments	To retain functionality of essential services



COMMON FIRE RATED COMPARTMENT

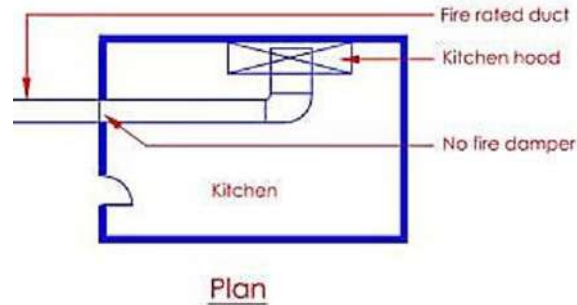
SPACE	SYSTEM	FD or FRD?	REMARK
SWITCHROOM	EAD,FAD	CONDITIONAL	To check if switch room comes with e-source. if yes, FRD, if no, FD
FIRE PUMP RM / GENERATOR RM	EAD,FAD	FRD	Ensure essential equipment remain functioning during e-mode. no FD allowed
FFL/SSL	FAD	FD	Fire damper to provide at each branch level, fad duct traversing from fan to first fire rated compartment shall be fire rated. no fd is allowed
STAIRCASE	FAD	CONDITIONAL FRD	No FD is required at each interval fag at common staircase void space. duct transverse out of staircase compartment to be fire rated. staircase more than 4 levels, interval fag to be provided.
UPS/BATTERY ROOM	FAD,EAD	FRD	Exhaust fan to be explosion proof
FLAMMABLE STORAGE ROOM	FAD,EAD	FRD	Duct traverse out of the compartment to be fire rated. no FD allowed.
FCC	FAD,SAD	FRD	FRD to be provided for essential ducting traverse out of FCC
SERVER ROOM	SAD,RAD,EAD	FD	EAD to provide motorised damper (normally tie with fire suppression system)
SMOKE PURGING / ESCS	FAD,EAD	FRD	To fire rate duct traverse out of compartment & within building edge



KITCHEN EXHAUST SYSTEM

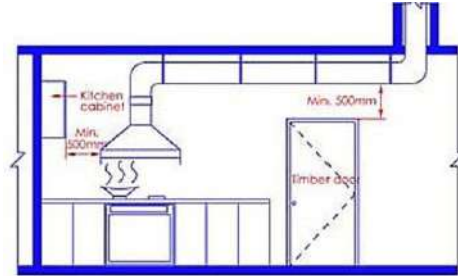
No fire damper required for kitchen exhaust duct due to

1. Grease on downstream would likely infinite before damper is closed
2. Fire damper may be falsely triggered due to high temperature of the exhaust air.



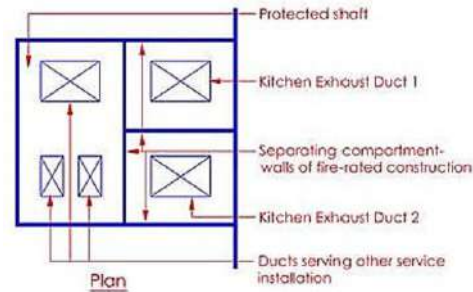
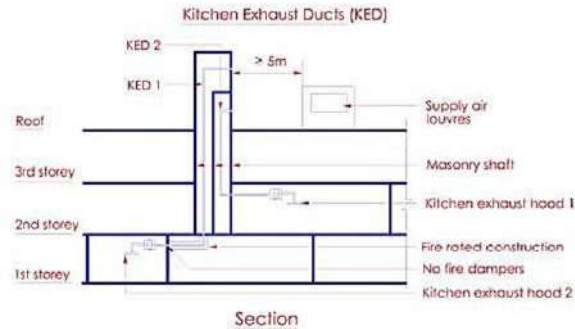
KITCHEN EXHAUST SYSTEM

1. KED traverse out of kitchen shall be fire rated (until building edge).
2. Hood and ductwork have to be 500mm away from combustible material.



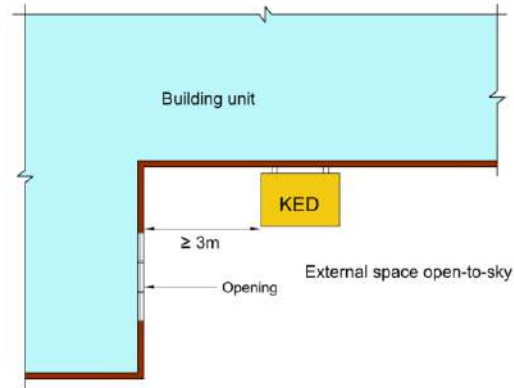
KITCHEN EXHAUST SYSTEM

1. KED passing through one or more floor slab shall be contained in protected shaft.
2. KEDs for separate hoods in separate compartments shall be separately contained with individual protected shaft.



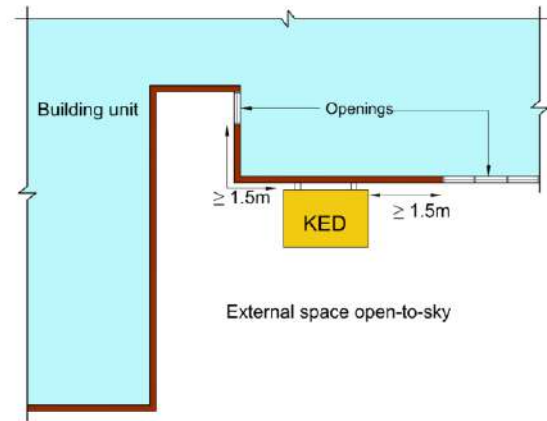
KITCHEN EXHAUST SYSTEM

KED running outside of building unit shall be $\geq 3\text{m}$ away from unprotected openings



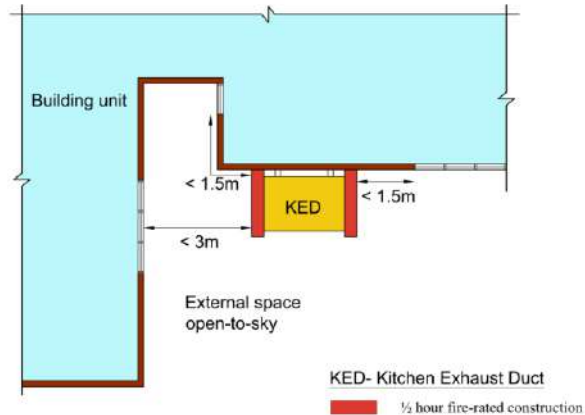
KITCHEN EXHAUST SYSTEM

If on same plane, offset requirement can be reduced by half to 1.5m.



KITCHEN EXHAUST SYSTEM

If setbacks cannot be met, fire rated construction has to be provided (at least 0.5hr rating) in between KED and unprotected openings.



PROVISION OF DUCT ACCESSORIES FOR CONTROL

Typical duct involved in system control.

1. Motorised damper (On-off/Modulating)
2. Non-return damper
3. Volume control damper



PROVISION OF DUCT ACCESSORIES FOR CONTROL

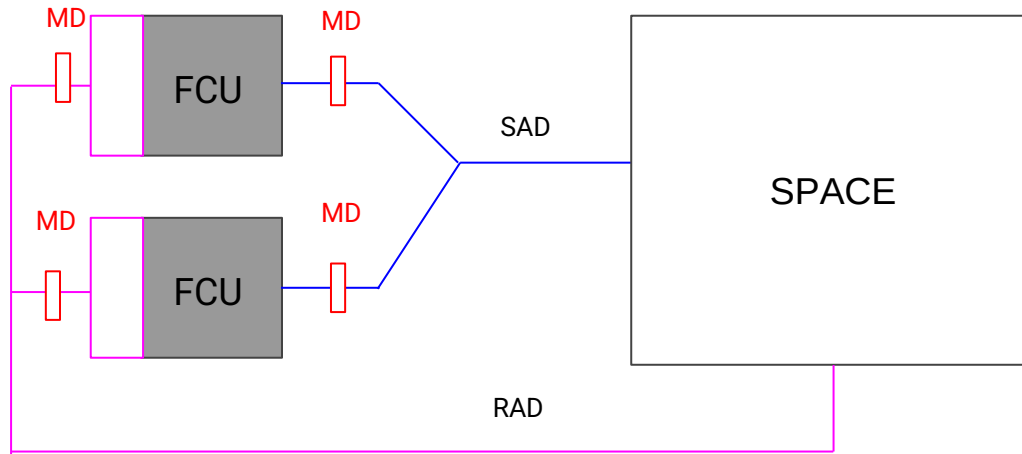
Understanding the provision in the context of

1. Functionality
2. System Control
3. Maintenance



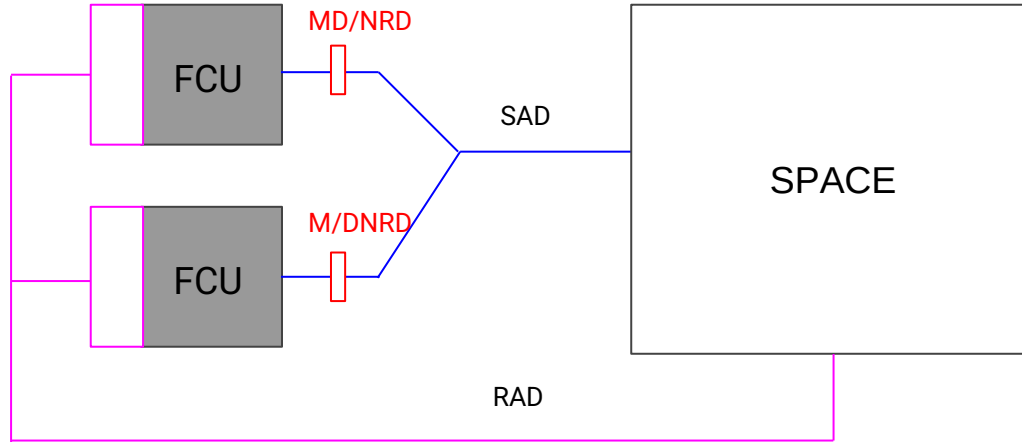
PROVISION OF DUCT ACCESSORIES FOR CONTROL

FCU (DUTY & STANDBY)



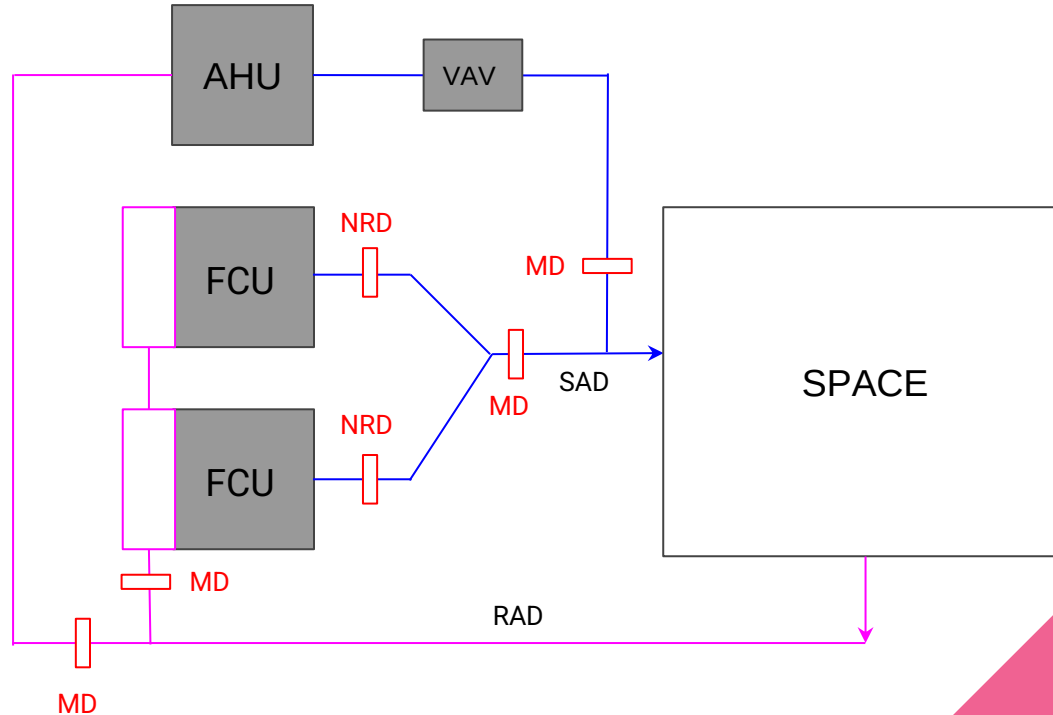
PROVISION OF DUCT ACCESSORIES FOR CONTROL

FCU (DUTY & STANDBY)



PROVISION OF DUCT ACCESSORIES FOR CONTROL

AHU & FCU



PROVISION OF DUCT ACCESSORIES FOR CONTROL

